Conserv Genet (2008) 9:531–546
DOI 10.1007/s10592-007-9367-2

RESEARCH ARTICLE

Population structure and genetic diversity of black redhorse (*Moxostoma duquesnei*) in a highly fragmented watershed

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Received: 12 January 2007 / Accepted: 12 June 2007 / Published online: 7 July 2007
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Abstract Dams have the potential to affect population size and connectivity, reduce genetic diversity, and increase genetic differences among isolated riverine fish populations. Previous research has reported adverse effects on the distribution and demographics of black redhorse (*Moxostoma duquesnei*), a threatened fish species in Canada. However, effects on genetic diversity and population structure are unknown. We used microsatellite DNA markers to assess the number of genetic populations in the Grand River (Ontario) and to test whether dams have resulted in a loss of genetic diversity and increased genetic differentiation among populations. Three hundred and seventy-seven individuals from eight Grand River sites were genotyped at eight microsatellite loci. Measures of genetic diversity were moderately high and not significantly different among populations; strong evidence of recent population bottlenecks was not detected. Pairwise F_{ST} and exact tests identified weak (global F_{ST} = 0.011) but statistically significant population structure, although little population structuring was detected using either genetic distances or an individual-based clustering method. Neither geographic distance nor the number of intervening dams were correlated with pairwise differences among populations. Tests for regional equilibrium indicate that Grand River populations were either in equilibrium between gene flow and genetic drift or that gene flow is more influential than drift. While studies on other species have identified strong dam-related effects on genetic diversity and population structure, this study suggests that barrier permeability, river fragment length and the ecological characteristics of affected species can counterbalance dam-related effects.

Keywords Dams · Habitat fragmentation · Moxostoma · Genetic diversity · Population structure

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Introduction

Redhorse suckers (*Moxostoma* sp.) are large-bodied benthic fishes that are typically found in rivers and larger streams (Jenkins and Burkhead 1993). The black redhorse (*Moxostoma duquesnei*) is one of six redhorse species in Ontario. It is generally found in medium-sized watercourses with well-developed pools and coarse, river bed materials with low levels of fine sediments (Bowman 1970; Kott et al. 1979). In Canada, black redhorse has only been collected from seven southwestern Ontario watersheds (Committee on the Status of Endangered Wildlife in Canada, COSEWIC 2005). Of these watersheds, it is believed extirpated from Catfish Creek and only widespread in the Grand River and Thames River watersheds. In Canada, black redhorse was designated as a threatened species by COSEWIC (2006). It is also considered imperiled in several neighbouring Great Lakes states (NatureServe 2005). Potential threats to black redhorse in Canada include high turbidity and siltation rates, high

nutrient levels, altered flow regimes, barriers to movement and physical habitat degradation (COSEWIC 2005; Portt et al. 2006).

Like many rivers in the Great Lakes basin, the Grand River watershed is fragmented by dams. Approximately 136 public and private dams have been identified throughout the watershed (W. Yerex, Grand River Conservation Authority, pers. comm.), including eight major dams operated to control flooding and augment low flows. Blockage of migratory routes by dams and the conversion of free-flowing rivers to reservoirs with poor tailwater habitat have been implicated in the decline of black redhorse populations in the United States (Santucci et al. 2005). Multi-species recovery strategies (Portt et al. 2006; TRRT 2005) and the recent COSEWIC status update identify dams as a threat to black redhorse populations. Other stresses to Grand River fish species at risk include impairment of water quality and quantity, baitfish harvesting, drain maintenance activities, habitat loss and degradation, and non-native species (Portt et al. 2006). In the Grand River watershed, black redhorse are absent upstream of major barriers along the upper reaches of the Conestogo River and Grand River, and completely absent from the highly fragmented Speed River sub-watershed (Reid et al. 2006). Its distribution is currently restricted to the less fragmented middle and lower areas of the watershed.

While previous studies have reported changes in distribution, growth and abundance (Patriarche and Campbell 1958; Quinn and Kwak 2003), the effects of dams on the genetic characteristics of black redhorse populations are unknown. As a result of changes to population size and gene flow, dams can reduce the genetic diversity within isolated fish populations and increase genetic differentiation among fish populations (Neraas and Spruell 2001; Meldgaard et al. 2003; Yamamoto et al. 2004). In isolated populations, low genetic diversity can result from genetic drift, inbreeding, bottlenecks and founder effects. Low genetic diversity can negatively affect a population's ability to respond to environmental changes, and increased inbreeding can expose deleterious recessive alleles that may reduce population fitness (Allendorf and Ryman 2002; Frankham et al. 2002). At larger spatial scales, by restricting the movements of individuals, dams can increase genetic differentiation between previously connected or panmictic fish populations (Meldgaard et al. 2003), create upstream–downstream gradients in genetic diversity (Yamamoto et al. 2004), and disrupt the regional equilibrium between drift and immigration of new alleles in populations (Hutchinson and Templeton 1999).

Future listing of black redhorse under Schedule 1 of the Canadian *Species at Risk Act* (SARA) will require the development of a recovery strategy and associated action plans. To identify actions that will protect genetic diversity and ensure population persistence (e.g. dam removal or construction of fish passage), it is necessary to describe the genetic characteristics of Grand River black redhorse populations, especially as it relates to the effects of dams. Secondly, the recent COSEWIC status assessment assumed that dams represented boundaries between black redhorse populations within watersheds. Future monitoring programs and status assessments require this assumption to be evaluated.

Using microsatellite DNA data, we addressed these information needs by describing the genetic structure and diversity of black redhorse in the Grand River watershed. The effect of fragmentation on black redhorse populations was evaluated by testing for evidence of: (1) population subdivision; (2) recent bottlenecks in isolated populations; (3) an additive effect of dams on population differentiation; and (4) a greater influence of drift, as compared to gene flow, on regional population structure.

Methods

Sample collection

During the spring (May–June) and late summer-early fall (late August to early October) of 2002 and 2003, black redhorse were sampled at eight sites (putative populations) in the Grand River watershed (43°21′N, 80°18′W) (Fig. 1). Site selection was based on the distribution of black redhorse and dams across the watershed. A recent inventory identified the distribution of black redhorse to be limited to seven river fragments. As only a single juvenile was captured from five sites along the Conestogo River between the St. Jacob's and Conestogo dams, black redhorse from this river fragment were not included in the study. Half of the sampled populations were separated by upstream and downstream dams. The five dams separating populations are between 1.5 m and 4.3 m high. Two (Caledonia weir and Mannheim weir) are equipped with Denil fishways. Two population pairs (PAR and NR; WM and CON) are not separated by dams, but instead represent mainstem and tributary populations.

Depending on watercourse size, black redhorse were collected using either backpack electrofisher, or a 5 KW pulsed DC boat-mounted electrofisher with a single boom anode. Along with the collection of fin clips for genetic analysis, total length was measured and aging structures (first pectoral fin ray) were collected before release. Fin clips were either stored in 95% ethanol or as dried tissue in scale sample envelopes. Ages were interpreted from sections of the left pectoral fin ray. Fin rays were embedded in epoxy, sectioned with a low-speed Isomet saw (section width 600 μm) mounted on a slide and polished with 320

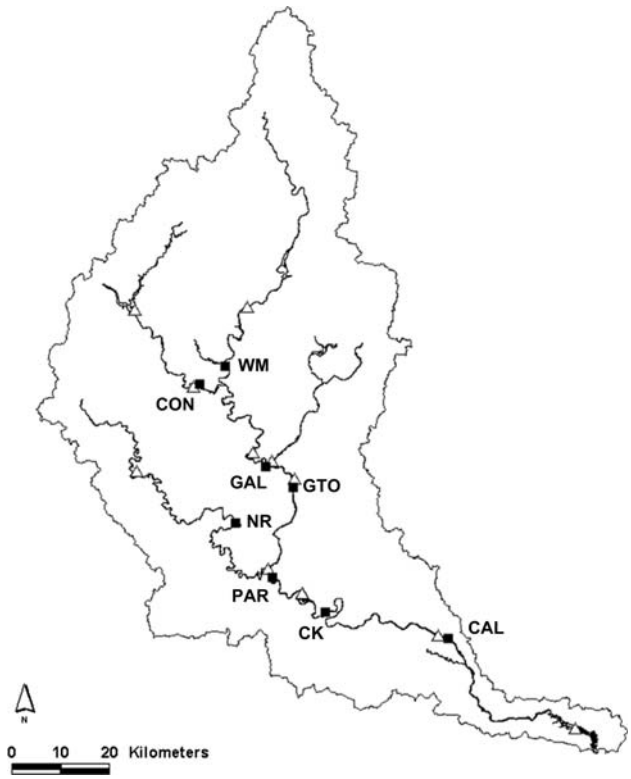

Fig. 1 Grand River sample sites (■) and dams (△) fragmenting black redhorse populations. Notation: Caledonia (CAL); Cockshutt (CK); Conestogo River (CON); Galt (GAL); GTO; Nith River (NR); Paris (PAR); and West Montrose (WM)

and 600 grit wet sandpaper. Fin ray sections were interpreted using a dissecting microscope.

Laboratory methods

Fin clips were lysed in a 96 deep-well plate with 400 µl of lysis buffer (50 mM Tris ph 8, 100 mM NaCl, 1 mM EDTA, 1% SDS w/v) and 1.5 µl Proteinase K (20 mg/ml). To precipitate DNA, 16 µl of 5 M NaCl and 800 µl of 80% isopropanol were added to each well. Samples were centrifuged for 45 min at 6,200 rpm to pellet DNA. Supernatant was removed and 1,000 µl of 70% ethanol was added to remove excess salts. Plates were vortexed and re-centrifuged for 45 min at 6,200 rpm. Ethanol was removed and DNA was resuspended in 100–200 µl of 1X TE (10 mM Tris, 1 mM EDTA).

Eight microsatellite DNA loci were amplified using the polymerase chain reaction (PCR) (Table 1). Microsatellite primers were initially developed for Lost River sucker (*Deltistes luxatus*) (Tranah et al. 2001), copper redhorse (*M. hubbsi*) (Lippe et al. 2004) and robust redhorse (*M. robustum*) (unpublished primers I. Wirgin, New York University, Tuxedo NY). Twenty-eight primers developed for these species were tested for suitability. Suitability was based on ability to amplify and level of polymorphism among individuals. For each primer, 4–5 samples were screened. Cross-reactivity (i.e. primer-dimer formation) of individual primer pairs was evaluated using AutoDimer (Vallone and Butler 2004).

Loci were amplified singly or in multiplex reactions (simultaneous amplification of two or three loci in one reaction). For subsequent detection, each microsatellite locus was amplified using an unlabelled and a fluorescent dye-labelled primer. PCR were performed in 11 µl total volumes containing: 1.5 µl of DNA template, 1.1 µl Thermocycler buffer, 0.8 µl of $MgCl_2$ [25 mM], 0.24 µl of dNTPs [10µM], 0.13 µl of *Taq* I DNA polymerase [5U/µl], microsatellite primers [10µM] (volumes listed below) and 5.6–6.0 µl of ddH_2O (depending on primer volume within each multiplex). *dlu*4296 (FAM) was amplified independently with a primer volume of 0.15 µl (PCR_1). In the first duplex reaction, primer volumes were 0.05 µl RR13 (FAM), 0.1 µl RR55 (NED) (PCR_2). In the second duplex reaction, primer volumes were 0.13 µl *Mohu-lav* 296 (FAM) and 0.10 µl *Mohu-lav* 305 (HEX) (PCR_3). In the triplex reaction, primer volumes were 0.10 µl *dlu*45 (FAM), 0.21 µl dlu405 (*HEX*) *and* 0.10 $l *Mohu-lav* 194 (NED) (PCR_4). Reactions were amplified in a MJ Research, Inc. PTC-100 thermocycler using an initial denaturation at 94°C (11 min), followed by 35 cycles of 94°C (1 min), 55°C (PCR_1 and PCR_2) or 60°C (PCR_3 and PCR_4) (1 min), and 72°C (1 min), with a final extension at 60°C (45 min).

PCR products were diluted with 11 µl of sterile deionized water and combined with an internal lane standard (formamide, loading buffer, and ROX 350 size standard (Applied Biosystems Inc.)). Diluted PCR products were denatured at 96°C for 2 min, loaded into a 5% Long Ranger acrylamide gel and electrophoresed on an ABIPrism 377 DNA sequencer. Reactions were visualized using GeneScan software and bands scored by comparison with standard ROX 350 ladder. Genotypes were checked for scoring errors attributable to stutter-products, large allele drop-out, or null alleles, using MICRO-CHECKER v2.2 (Van Oosterhout et al. 2004).

Data analysis

Patterns of allelic diversity and disequilibrium

The number of alleles (N_A), standardized allelic richness (A_R), expected and observed heterozygosity (H_E and H_O, respectively) statistics and allele frequencies were compiled using FSTAT version 2.9.3 (Goudet 2001) and the Microsoft excel add-in Microsatellite toolkit (Park 2001). Differences in A_R among sites and among loci were each tested using a one way analysis of variance (ANOVA).

Table 1 Details of microsatellite primers used to investigate population structure of black redhorse populations in the Grand River

Locus	Repeat motif	Primer sequences (5′–3′)	Allele size range (bp)	References
dlu45	$(GATA)_{29}$	F:TGGGCCTTAGTGCAGAGGA R:TGGTTAGGCAGAATTCTCCAG	273–337	Tranah et al. (2001)
dlu405	$(GATA)_{21}$	F:CAGCCCTCCGCGTGAAAACAAT R:ACCGTAAGGGGGCAGCAGAAGG	221–337	Tranah et al. (2001)
dlu4296	$(GATA)_{27}$	F:AAGAACAATTTAAAACAGTGAGTG R:TACCCTTATGTTTAATGTGTTAGG	153–253	Tranah et al. (2001)
Mohu-Lav 194	$(GATA)_{24}$	F:CTTTTTCTCCTGGCGAACG R:CACGCAGCGGAGGTATTATT	160–236	Lippe et al. (2004)
Mohu-Lav 296	$(ATCT)_{21}(TTCT)_{11}$	F:TCCTGCTATCTTTGGCATATTTT R:TGCCCAACAGAGAAAGGAAC	129–229	Lippe et al. (2004)
Mohu-Lav 305	$(GATA)_9(GACA)_3GAGA(GACA)_7(TAGA)_{19}$	F:TGGAGAGTTATTTTTCTCACATCTAA R:AACTTGAATGTTTGATATTGCTTTT	116–236	Lippe et al. (2004)
RR13	$(AC)_{13}$	F:AGGAGGGAATGAAGAATC R:TGAAAAATGGCAGGTGTC	191–269	Wirgin unpubl.
RR55	$(TG)_5(GT)_{48}$	F:CTTCTCATTGCCTTACAT R:TCCCCTTCTCCATCTTTG	108–190	Wirgin unpubl.

Conformation of A_R data to the assumption of normality was assessed with the Shapiro-Wilks test. Analysis of Molecular Variance (AMOVA) was completed using GENALEX 6 (Peakall and Smouse 2006). By facilitating downstream dispersal and limiting upstream dispersal, the unidirectional flow of water downstream in riverine systems has been hypothesized to create upstream–downstream gradients of genetic diversity (Hernandez-Martich and Smith 1997). Therefore, Pearson product moment correlations (r) were calculated between distance upstream from the mouth of the Grand River (rkm) and both A_R and H_O.

Genotypes at each locus for each population were tested for conformance to Hardy-Weinberg equilibrium (HWE) using GENEPOP version 3.4 (Raymond and Rousset 1995). Fisher exact test of linkage disequilibrium was performed between all pairs of loci with 10,000 dememorizations, 10,000 batches and 10,000 iterations per batch. To address the risk of inflated Type I error resulting from multiple tests, significance levels ($P < 0.05$) were adjusted using the standard Bonferroni correction.

To determine globally whether individual black redhorse collected from each site were more related than expected, given that their parents mated randomly, a Monte Carlo re-sampling test (IDENTIX, Belkhir et al. 2002) that uses a multilocus estimator ($r_{x,y}$) for pairwise comparisons (Queller and Goodnight 1989) was applied. High relatedness values and/or excess representation by a few families could be indicative of inbred individuals in a population (Lippe et al. 2006). Tests of both the mean and variance of relatedness coefficients were undertaken, as significantly higher variance can indicate that several independent groups of related individuals were sampled. As populations did not conform to Hardy-Weinberg expectations, re-sampling (1,000 permutations) was carried out at the genotype level.

Bottlenecked populations may exhibit gametic disequilibrium (Waples 2002), reduced genetic diversity (particularly the loss of rare alleles) (Allendorf 1986), and increased heterozygosity relative to that expected at mutation-drift equilibrium (Cornuet and Luikart 1996). Evidence of recent and pronounced reductions in population size was tested using BOTTLENECK 1.2.02 (Piry et al. 1999) and by calculating Garza and Williamson's (2001) M statistic. When a population undergoes a reduction in its effective population size, allelic diversity is reduced more rapidly than heterozygosity and, as a consequence, there is a transient deficiency in the number of alleles (Cornuet and Luikart 1996). BOTTLENECK compares a population's heterozygote excess to that which is expected to be found at mutation-draft equilibrium. A stepwise-mutation model (SMM) of microsatellite evolution was assumed and 5,000 simulation iterations were conducted for each population (Piry et al. 1999). Significance of heterozygote excess over all loci was tested with the Wilcoxon signed-rank test. Garza and Williamson's (2001) M is the mean of the ratio of the total number of alleles (k) and the number of repeats over the allele size range (r) for each loci. The ratio is based on the expectation that when the effective size of a population is reduced and drift occurs, reductions in allele size range will occur more slowly than the loss of individual alleles. Based on simulations and data from natural populations, values of

M < 0.68 are indicative of populations that have undergone a recent bottleneck (Garza and Williamson 2001).

Population structure

Differentiation among populations was evaluated using both population level (F_{ST} and exact tests) and individual-based clustering (STRUCTURE) approaches. Global F_{ST} and pairwise F_{ST} were estimated using Weir and Cockerham's Θ (Weir and Cockerham 1984). F_{ST} values and significance of estimates were calculated with FSTAT. F_{ST} was used instead of R_{ST}, as F_{ST} estimates are more conservative when relatively few (< 20) microsatellite loci are used and when populations have recently diverged (Gaggiotti et al. 1999). Exact tests (based on the distribution of alleles across populations) were performed for each locus and population pair using GENEPOP (genic differentiation option) (Raymond and Rousset 1995). Results of locus specific tests were combined (Fisher's Method) to give an overall test of differentiation among populations. This approach gives more weight to rare alleles and is considered to be more sensitive to the detection of weak population differentiation than F_{ST}.

Genetic distances among population pairs were estimated by computing Cavalli-Sforza and Edwards (1967) genetic chord distances (D_{CE}) in POPULATIONS version 1.2.28 (Oliver Langella 1999 unpublished, http://www.pge.cnrs-gif.fr/bioinfo/populations/index.php). D_{CE} was used because it is commonly applied in studies of fish population structure, does not assume any mutation model, and performs well in simulations with microsatellite data (Takezaki and Nei 1996). To visualize the genetic relationships among sites, D_{CE} chord distances were used to build an unrooted neighbour-joining tree in TREEVIEW version 1.66 (Page 1996). Robustness of the typology was evaluated by 1,000 bootstrap resamplings as implemented in POPULATIONS.

A Bayesian genotypic clustering method (STRUCTURE version 2.1) (Pritchard et al. 2000) was used to corroborate population-based approaches used to determine the number of black redhorse populations in the Grand River. STRUCTURE uses the assumption of HWE and linkage equilibrium within populations to identify the number of populations (K) that best fit the data; as well as assignments of individuals to populations such that deviations from HWE and linkage disequilibrium are minimized (Pearse and Crandall 2004). The most likely value of K was selected on the basis of maximizing ln (K) (Pritchard et al. 2000) and the rate of change in the log probability of data between successive K values (Evanno et al. 2005). Mean values of these two parameters were calculated from 20 simulations of each model (k = 1–9) assuming admixture and correlated allele frequencies between populations with 50,000 replications burn-in period and 50,000 Markov Chain Monte Carlo replicates. Due to heterozygote deficiency across most populations, two loci (*Mohu–Lav*296 and RR55) were not included in simulations.

The effect of dams on genetic differentiation was evaluated by testing for significant correlations between matrices of pairwise genetic distance (F_{ST} and D_{CE} values), geographic distance and number of dams separating population pairs. As geographic distances between population pairs were greater than habitat width, F_{ST} values were adjusted with the following transformation: $F_{ST}/(1–F_{ST})$ (Rousset 1997). Significance of correlations was based on Mantel and partial Mantel tests calculated using the software IBD (Bohonak 2002).

Two approaches were applied to assess whether black redhorse populations were in regional equilibrium (i.e. a balance between the loss of alleles due to drift and their replacement by gene flow). The first approach was based on the relationship between pairwise F_{ST} and geographic distances. Assuming a stepping stone model of regional population structure, Hutchison and Templeton (1999) interpreted populations of a region to be at equilibrium (Case I) if: (1) there is a significant relationship between pairwise F_{ST} genetic and geographic distances separating populations; and (2) a scatterplot of pairwise F_{ST} and geographic distances reveals a positive and monotonic relationship over all distance values of a region. Descriptions of cases II, III and IV where populations are not in regional equilibrium are provided by Hutchison and Templeton (1999). The second approach, using a Markov Chain Monte Carlo (MCMC) approach implemented in 2mod (Ciofi et al. 1999; software available from http://www.rubic.rdg.ac.uk/mab/software.html), calculated the relative likelihood of two alternate models of demographic history: a model of immigration-drift equilibrium versus a model of drift (without equilibrium). The MCMC search was carried out twice for 10^5 iterations with the initial 30,000 discarded as burn-in.

Results

Genetic diversity and disequilibrium

All loci were moderately to highly polymorphic, with the number of alleles per locus ranging from 12 to 33 (mean = 19.4), and observed heterozygosity (H_O) ranging from 0.60 to 0.67 (mean = 0.63) (Table 2). The mean number of private alleles per population was 4.1 (range: from 2 to 6). Allelic richness was significantly different among loci (Shapiro-Wilks: $P > 0.05$, ANOVA: F = 93.2, $P < 0.0001$) but not among Grand River sites (Shapiro-Wilks: $P > 0.05$, ANOVA: F = 0.03, $P = 0.99$). Ninety-eight percent of allelic diversity was attributed to individuals

Table 2 Genetic diversity at eight microsatellite loci for Grand River populations; includes total number of alleles (A), allelic richness (A_R) standardized at 22 individuals per population, mean expected heterozygosity (H_E), and observed heterozygosity (H_O)

Loci	A	A_R	H_E	H_O	A	A_R	H_E	H_O	A	A_R	H_E	H_O	A	A_R	H_E	H_O
		CAL (n = 51)				CK (n = 49)				PAR (n = 23)				NR (n = 47)		
dlu45	8	6.7	0.71	0.54	9	7.9	0.77	0.69	7	7.0	0.78	0.73	10	8.4	0.79	0.61
dlu405	18	13.6	0.91*	0.66	18	14.4	0.91	0.88	13	12.8	0.90*	0.61	18	15.7	0.93	0.94
dlu4296	18	14.8	0.93	0.84	18	14.2	0.92	0.96	16	15.7	0.92	0.83	19	15.7	0.92	0.89
Mohu-Lav194	10	7.9	0.83	0.76	12	5.7	0.80	0.67	9	5.9	0.86	0.83	10	8.3	0.85	0.77
Mohu-Lav296	20	8.8	0.93*	0.56	13	9.8	0.81	0.82	19	9.0	0.95*	0.73	16	9.1	0.90*	0.43
Mohu-Lav305	10	17.3	0.60	0.44	8	12.4	0.46	0.33	9	20.0	0.49	0.39	9	13.5	0.66*	0.43
RR13	6	4.0	0.47	0.57	4	2.9	0.49	0.60	3	3.0	0.57	0.74	5	3.4	0.52	0.57
RR55	8	6.2	0.57*	0.37	8	5.9	0.64*	0.37	5	5.0	0.63	0.39	7	5.8	0.45*	0.26
Mean	12.3	9.9	0.74	0.59	11.3	9.2	0.73	0.67	10.1	9.8	0.76	0.66	11.8	10.0	0.75	0.61
		GTO (n = 49)				GAL (n = 52)				CON (n = 51)				WM (n = 55)		
dlu45	8	8.4	0.64	0.61	10	8.4	0.80	0.80	9	7.5	0.76	0.64	10	7	0.76	0.64
dlu405	20	14.9	0.93	0.94	19	14.9	0.91*	0.67	22	17.3	0.94	0.9	20	12.8	0.93	0.98
dlu4296	18	13.8	0.92	0.98	17	13.8	0.92	0.92	20	16.4	0.93	0.92	20	15.7	0.92	0.89
Mohu-Lav194	10	5.1	0.59	0.59	11	5.1	0.80	0.65	12	5.1	0.76	0.75	10	5.9	0.82	0.73
Mohu-Lav296	16	9.8	0.90*	0.69	18	9.8	0.92*	0.43	17	9.9	0.88*	0.69	16	9	0.90*	0.46
Mohu-Lav305	6	14.3	0.49	0.38	6	14.3	0.54*	0.31	6	13.7	0.44	0.39	10	20	0.57	0.45
RR13	3	3.4	0.55	0.58	4	3.4	0.46	0.52	5	3.5	0.49	0.45	4	3	0.51	0.36
RR55	11	6.5	0.64	0.41	10	6.5	0.62	0.46	6	4.3	0.55*	0.35	12	5	0.67*	0.49
Mean	11.5	9.5	0.71	0.65	11.9	9.5	0.75	0.60	12.1	9.7	0.72	0.64	12.8	9.8	0.76	0.63

For each population, loci that deviate from HWE are identified by superscript: *

within populations (AMOVA: $P < 0.0001$) with only 2% of allelic diversity found among populations. No longitudinal gradient in genetic diversity along the Grand River was detected (A_R: r = 0.12, P = 0.78; H_O: r = 0.25, P = 0.56). For each microsatellite locus and population, allele size frequencies are presented in Appendix I.

Tests of heterozygote deficiency showed deviations from HWE in 17 of 64 (27%) population locus combinations: seven at *Mohu-Lav*296, five at RR55, three at *dlu*405 and two at *Mohu-Lav*305. Linkage disequilibrium was not detected at any of the 224 combinations of loci pairs over eight collection sites. Across all populations, linkage disequilibrium was detected at only one of the 28 loci pairs (*Mohu-Lav*194 and *Mohu-Lav*305). A significantly greater level of relatedness among individuals than expected from a random population sample was only detected for black redhorse collected from NR (mean $r_{x,y}$ = –0.01406: P = 0.003; variance $r_{x,y}$ = 0.03316: P = 0.008).

Strong evidence of recent population bottlenecks was not detected. For all eight populations, Wilcoxon signed-rank tests for heterozygote excess were not significant (P-values: 0.84–0.99); indicating no significant departure from mutation-drift equilibrium. For all populations except CK (M = 0.60), M was greater (M: 0.74–0.91) than the threshold value (M < 0.68) proposed by Garza and Williamson (2001).

Population structure

The global F_{ST} value for the eight populations was low: 0.011. Despite low pairwise F_{ST} values (range: 0.001–0.032), significant differences were found for 17 of 28 (61%) pairwise population comparisons (Table 3). Significant pairwise differences existed for all neighbouring populations separated by dams without fishways. Pairwise differences were not as consistent for population pairs that were either not separated by dams (1 of 2 pairs) or separated by dams with fishways (2 of 3 pairs). Exact tests detected a greater amount of population subdivision than pairwise F_{ST} comparisons; 25 of 28 (89%) pairwise comparisons were significantly different (Table 3). All neighbouring populations separated by dams (with and without fishways) were significantly different. Despite statistically significant evidence against panmixia, low pairwise F_{ST} values and the low number of significant single locus tests (less than 2 of 8 loci for 25 of 28 comparisons) indicate weak levels of population subdivision.

The neighbour joining dendrogram (not shown) suggests less population subdivision than pairwise F_{ST} and exact tests. Based on bootstrap values greater than 50%, GTO, GAL and CON were clustered together separate from other populations. Results from STRUCTURE indicated that two population groups exist in the Grand River watershed (Fig. 2). However, no meaningful interpretation of population structure could be made as individuals from each of the eight populations were equally distributed between the two population groups.

Genetic distances between population pairs were not affected by either geographic distance or the number of intervening dams. Pairwise F_{ST} and D_{CE} genetic distances were not correlated to geographic distance (partial Mantel: F_{ST}: Z = 20.1, r = –0.16, 1-sided P = 0.72; D_{CE}: Z = 475.9, r = –0.23, 1-sided P = 0.82) or number of intervening dams (partial Mantel: F_{ST}: Z = 1.3, r = 0.28, 1-sided P = 0.07; D_{CE}: Z = 26.3, r = –0.097, 1-sided P = 0.28).

The scatterplot of F_{ST} values against geographic distance (Fig. 3) suggests that Grand River populations are not in regional equilibrium and that gene flow is more influential than drift (Case II, Hutchinson and Templeton 1999). In all iterations of the two runs (70,000 per run), I (parameter estimated by 2mod) indicated that a migration-drift equilibrium model was more likely than a non-equilibrium drift only model.

Discussion

While other studies have identified strong dam-related effects on the genetic diversity and population structure of other riverine fishes, similar effects were not detected for black redhorse populations in the highly fragmented Grand River. Measures of genetic diversity were moderately high and not significantly different among populations. Population differentiation was weak across the watershed and not related to either geographic distance or the number of intervening dams. Tests for regional equilibrium indicated that populations were either in equilibrium or that gene flow is more influential than drift.

Genetic diversity

Fragmentation of riverine fish populations by dams can lower genetic diversity by limiting gene flow between populations and reducing effective population size (Jager et al. 2001). Relative to below dam populations, lower genetic diversity has been measured in headwater populations of river sculpin (*Cottus gobio*) (Hanfling and Weetman 2006) and white-spotted char (*Salvelinus leucomaenis*) isolated by dams (Yamamoto et al. 2004). For both species, a lack of upstream gene flow to balance genetic drift (and the associated loss of alleles) was the most likely cause. Alternatively, no differences were found between the genetic diversity of above- and below-dam populations of bull trout (*S. confluentus*) in Clark Fork River system, Montana (Neraas and Spruell 2001); steelhead trout (*Oncorhynchus mykiss*) in the Russian River, California

Table 3 Summary of pairwise tests of genetic differentiation among Grand River black redhorse populations

	CAL	CK	PAR	NR	GTO	GAL	CON	WM
CAL		0.008*	0.008*	0.009	0.021	0.007*	0.012	0.001^{NS}
CK	2		0.008	0.017	0.014	0.004	0.009	0.006*
PAR	2	1		0.002^{NS}	0.021	0.009^{NS}	0.013	0.001*
NR	1	4	1*		0.032	0.018	0.018	0.007^{NS}
GTO	5	2	2	2		0.016	0.007*	0.014
GAL	1	1	3	2	2		0.004*	0.011
CON	3	2	1	1	1	1		0.008
WM	1*	1	1	0*	2	1	1	

Pairwise F_{ST} estimates are presented above the diagonal. The number of significant locus-population pair exact tests (genic differentiation option, GENEPOP) is presented below the diagonal. Non-significant pairwise differences (after Bonferroni correction) for F_{ST} estimates and exact tests (all loci combined, Fisher's Method) are indicated by superscripts: * ($P < 0.05$) and NS ($P > 0.05$)

(Denier et al. 2007); and brown trout (*Salmo trutta*) along the Måna River, Norway (Heggenes and Roed 2006). In the case of Russian River steelhead, the lack of an above-below dam population difference was proposed to be due to the relatively short period of isolation (< 45 years), the movement of adults past the dam, and the relatively large amount of habitat still available to isolated upstream populations (Denier et al. 2007).

The lack of difference in genetic diversity among Grand River black redhorse populations may be the result of dams being permeable to upstream and downstream migration, the length of river fragments and the relatively long life-span of black redhorse. At the Mannheim weir and the downstream Dunnville dam, various redhorse species pass upstream through Denil fishways during spring spawning migrations (Bunt et al. 2001; Reid 2006). The Wilkes dam, which separates black redhorse at CK from PR and NR, is only 1.5 m high and therefore likely passable by redhorse species during spring high flow periods. During spawning runs, steelhead trout are able to swim from Lake Erie up the Grand River to Paris and further along the Nith River to the town of New Hamburg.

For some riverine fishes including redhorses, the creation of small river fragments by dams has resulted in small population sizes (Jager et al. 2001; Reid unpublished data). Small populations are most vulnerable to the loss of alleles through genetic drift, and are often characterized by low levels of genetic variation (Frankham 1996). Assuming habitat size is a surrogate for population size, the genetic structure of populations in small river fragments would change more rapidly then populations in large river fragments or unfragmented rivers. However, river fragments supporting Grand River populations of black redhorse are relatively large (18–103 km long) and, therefore, unlikely to negatively affect population size. Further, tests for bottlenecks do not indicate that recent population declines and concurrent losses of genetic diversity have occurred.

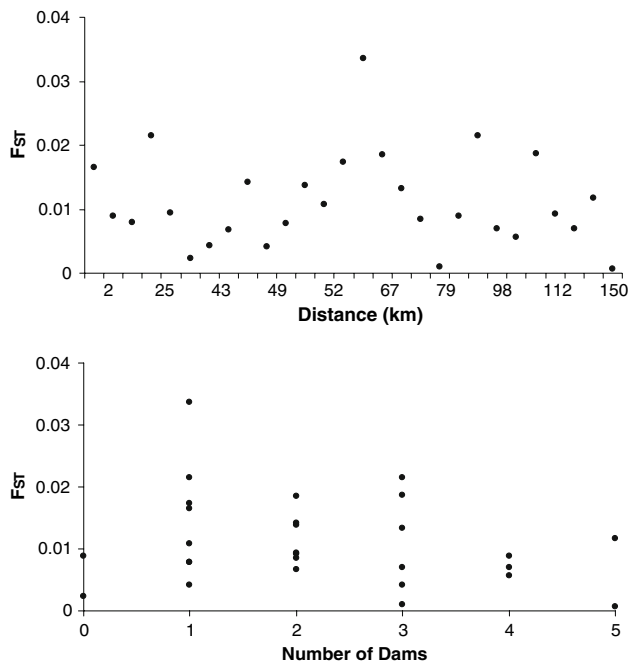

Fig. 3 Scatterplots of pairwise F_{ST} values versus geographic distance (top) and versus number of dams (bottom) separating pairs of Grand River black redhorse populations

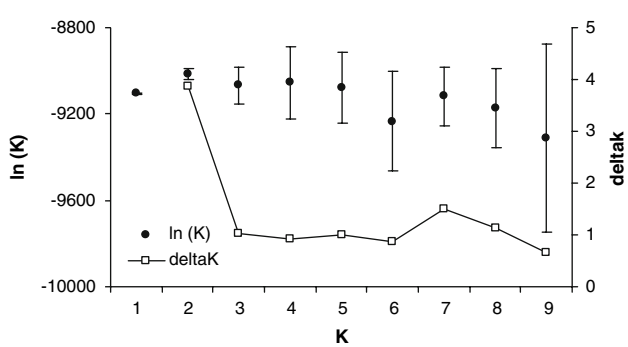

Fig. 2 Comparison of ln (K) and deltaK values calculated from STRUCTURE output where the hypothesized number of populations (K) ranged from 1 to 9

The absence of expected declines in genetic diversity in population studies of catostomids has been attributed to their relatively late age of maturity and long life-spans (Whitehead et al. 2003; Lippe et al. 2006). In the Grand River, black redhorse begin to mature at age 4 and live up to at least 16 years (Reid 2006, Table 4). Therefore, compared to fishes with shorter generation times and life-spans (e.g. river sculpin), black redhorse would be less vulnerable to rapid loss of genetic diversity resulting from population declines and isolation. Heterozygosity (H_O) of Grand River black redhorse populations was lower than that measured for copper redhorse, river redhorse (*M. carinatum*) and shorthead redhorse (*M. macrolepitodum*) in other rivers of the Great Lakes-St. Lawrence basin (Lippe et al. 2006; Reid unpublished data). However, it was still greater than that reported by DeWoody and Avise (2000) for most freshwater fishes.

Heterozygote deficiency can result from the occurrence of null alleles, inbreeding, and/or population admixture (Wahlund effect). Null alleles are alleles that do not amplify during PCR which can result in either no PCR product or in a false homozygote individual, if the locus is heterozygote. Across each population sample, MICROCHECKER identified the potential for null alleles at two to five loci. This was only relatively consistent for RR55 and *Mohu-Lav* 296. However, the frequency of non-amplified samples was low for all loci (mean: 0.6%; range: 0–3%). It is also unlikely that up to five loci show null alleles simultaneously. As multiple age classes of black redhorse were present in all samples (Table 4), observed heterozygote deficiency would more likely be the result of population admixture.

Population genetic structure

River fragmentation can also increase genetic differentiation among populations, as measured for brown trout (Heggenes and Roed 2006), European grayling (*Thymallus thymallus*) (Meldgaard et al. 2003), river sculpin (Hanfling and Weetman 2006) and white-spotted charr (Yamamoto et al. 2004). While some differentiation among Grand River black redhorse populations exists, the influence of fragmentation on genetic differentiation was limited. F_{ST} values for population pairs separated by dams were much lower than that reported for fragmented salmonid (Meldgaard et al. 2003; Yamamoto et al. 2004; Heggenes and Roed 2006) and river sculpin populations (Hanfling and Weetman 2006). Additionally, pairwise genetic distance estimates were not correlated with either geographic distance or the number of intervening dams.

As discussed for genetic diversity, the lack of a strong dam related effect may be explained by the nature of habitat fragmentation in the Grand River. Increased genetic divergence between populations fragmented by dams has been attributed to the additive negative effect of multiple barriers on gene flow (Meldgaard et al. 2003; Hanfling and Weetman 2006), and the reduction in effective population size and increase in genetic drift associated with small, isolated populations (Jager et al. 2001; Hanfling and Weetman 2006). Low F_{ST} values and the use of fishways by redhorse species suggest that dams along the Grand River are likely permeable barriers. Empirical research by Yamamoto et al. (2004) and population modelling by Jager et al. (2001) have found a negative relationship between genetic differentiation and the amount of habitat available to isolated populations. Small isolated headwater fish populations are vulnerable to the loss of alleles as a result a genetic drift and population bottlenecks (Yamamoto et al. 2004; Hanfling and Weetman 2006). Alternatively, river fragments in the Grand River are relatively large, support black redhorse populations that have not experienced recent bottlenecks and (if habitat availability is surrogate for population size) likely have large effective population sizes. Although no population estimates are available, black redhorse were commonly captured during electrofishing surveys within their Grand River distribution.

Ecological and life history traits can also influence population genetic divergence through their effects on genetic drift and gene flow. Variation in population structuring has been attributed to species differences in spawning ecology, dispersal potential and effective population size (Castric and Bernatchez 2004; Whiteley et al. 2004). Considering these factors, black redhorse are not likely to show strong population structure within watersheds. The high degree of structuring between populations of bull trout, as compared to mountain whitefish (*Prosopium williamsoni*), in the Clark Fork River, Montana, was explained in terms of the more extensive migrations to spawning sites, high fidelity to natal spawning sites, mate selection and smaller effective populations sizes (Whiteley et al. 2004). Black redhorse are group spawners that do not demonstrate mate choice or spawning site fidelity

Table 4 Total length (TL) and age statistics for black redhorse sampled from the Grand River

Site	TL (cm)		Age (years)	
	Mean	Range	Mean	Range
CAL	37.3	9.8–46.2	5.4	0–15
CK	33.8	16.8–44.4	4.6	1–10
PAR	46.3	38.0–50.8	9.7	5–14
NR	35.5	16.5–49.3	6.2	1–14
GTO	35.6	25.0–48.8	6.5	4–12
GAL	31.9	7.4–51.6	4.7	0–15
CON	28.8	14.4–52.1	4.3	1–15
WM	22.3	9.8–51.1	3.1	1–16

(Bowman 1970; Kwak and Skelly 1992). River sculpin, which were strongly effected by weirs, typically move less than 100 m within a given year (Knaepkens et al. 2004) and have low levels of fecundity (Hanfling and Weetman 2006). In contrast, black redhorse undertake comparatively large seasonal migrations to spawning and over-wintering habitats (Bowman 1970; Smith 1977) and are relatively fecund. Egg numbers from mature females from the Grand River have been reported between 4,126 and 11,551 (Kott and Rathmann 1985).

The absence of an isolation by distance effect could reflect a lack of conformity of data with test assumptions or limitations associated with the test. It may be unrealistic to assume that Grand River black redhorse populations are in equilibrium (Slatkin 1993), conform to the stepping-stone model of population structure (Taylor et al. 2003), or that dispersal patterns translate unequivocably into spatial patterns of genetic diversity (Castric and Bernatchez 2004). Tests for regional equilibrium indicate that Grand River black redhorse populations were either in equilibrium between gene flow and genetic drift (migration-drift equilibrium model; Ciofi et al. 1999) or that gene flow is more influential than drift (CASE II; Hutchinson and Templeton 1999). The absence of a relationship between genetic and geographic distances, combined with low variance in estimates of divergence, is indicative of new populations with a recent evolutionary history of colonizing from a common homogeneous source (Hutchinson and Templeton 1999). Therefore, a sufficient number of generations may not have passed since colonization after glacial retreat for Grand River populations to have reached regional equilibrium (Crispo and Hendry 2005). At the spatial scale of the Grand River watershed, low spawning site fidelity and high dispersal potential suggest that gene flow between black redhorse populations is not likely restricted to adjacent populations, as assumed in the stepping-stone model of population structure (Hedrick 2000). Castric and Bernatchez (2004) found individual assignment tests to be more powerful in describing spatial patterns of dispersal in St. Lawrence River tributary populations of Atlantic salmon (*S. salar*) and brook trout (*S. fontinalis*) than traditional IBD approaches. Unfortunately, the moderate to high levels of gene flow between Grand River populations and number of loci genotyped limit the utility of individual based-methods in this study.

Despite evidence of population structuring across the Grand River watershed provided by population-based methods (e.g. pairwise F_{ST} comparisons), STRUCTURE was unable to differentiate any of the populations. A similar lack of structure was found using STRUCTURE on copper redhorse samples collected from the St. Lawrence River and Richelieu River in Quebec (Lippe et al. 2006). The ability of STRUCTURE to correctly identify the number of populations is dependent on the level of gene flow, the number of individuals sampled within each population and the number of genotyped loci (Waples and Gaggiotti 2006). Global F_{ST} estimates for both black redhorse ($\Theta = 0.011$) and copper redhorse ($\Theta = 0.0038$) were low, suggesting either high levels of gene flow or recent divergence of sampled populations. Using simulated datasets, Latch et al. (2006) found that STRUCTURE could not detect more than one population at $F_{ST} = 0.01$ and could only correctly identify the number of populations when $F_{ST} > 0.03$. Similarly, Waples and Gaggiotti (2006) found STRUCTURE to perform poorly when $F_{ST} \sim 0.01$. While the number of loci and number of individuals from each population (except Paris) genotyped in this study should provide sufficient power for genetic distance and population structure measures (Ruzzante 1997; Ryman et al. 2006), when gene flow is moderate or high, it maybe insufficient for Bayesian assignment methods such as STRUCTURE (Waples and Gaggiotti 2006). This explanation, however, cannot be used to interpret the results of Lippe et al. (2006), who genotyped 22 microsatellite loci.

Conservation implications

Quantitative assessment criteria applied by COSEWIC to determine whether a species is considered to be threatened or endangered include the number of locations where it currently exists, and whether declines or extreme fluctuations in the number of locations are occurring (COSEWIC 2004). Location is defined as a geographically distinct area where a group of individuals of a species is present and where dispersal between locations is impossible or very rare. In the recent COSEWIC black redhorse status update, the number of locations identified in the Grand River (7) and Thames River (5) was based on the assumption that dams represent strong barriers to dispersal (COSEWIC 2005). Results from this study indicate weak population structure in the Grand River and that dams have had a limited influence on that structure. In the Thames River, recent telemetry monitoring tracked the upstream passage of shorthead redhorse and white sucker (*Catostomus commersonii*) through Springbank dam (one of the assumed barriers to black redhorse dispersal) when stop-log gates are not in place (November to late May) (Biotactic 2006). Therefore, weak Grand River population structure and observations from the Thames River indicate that dams do not represent strong barriers to dispersal and the number of black redhorse locations was overestimated. Future COSEWIC status assessments of this relatively mobile species require a determination of whether the term "location" should be applied to populations within watersheds or whether it should be restricted to the individual south-western Ontario watersheds where it is found.

Characterization of the genetic structure of black redhorse populations throughout southwestern Ontario would assist in this determination.

Relatively high levels of genetic diversity and gene flow, and the presence of multiple year classes at collection sites, indicate that Grand River black redhorse populations are currently not in decline. Past studies identifying a negative effect on black redhorse generally investigated large hydro-electric and flood control dams which, in addition to having large upstream impoundments, created downstream tailwater habitats characterized by releases of cold hypolimnetic water and variable flows (Patriarche and Campbell 1958; Eley et al. 1981; Swink and Jacobs 1983; Quinn and Kwak 2003). Peaking based hydro-power generation does not occur in the Grand River and most dams separating black redhorse populations are low-head weirs that do not create large upstream impoundments. While past research has identified strong effects on riverine fish genetic diversity and population structure, this study suggests the nature of river fragmentation and the ecological characteristics of affected species can have strong mitigating influences on dam-related effects. This interpretation is consistent with recent reviews of habitat fragmentation studies for other taxa (Debinski and Holt 2000; Ewers and Didham 2006), and recognition that dam characteristics such as size and operational mode have a strong influence on impacts (Poff and Hart 2002).

Anticipated future urban growth will increase demand for water, and climate change is expected to reduce groundwater supplies and mean annual flow levels in the Grand River (Southam et al. 1999). These stresses will likely have a negative effect on the quantity and quality of black redhorse habitat and, therefore, species recovery. While the current practise of augmenting summer low flows with releases from upstream reservoirs is likely beneficial to black redhorse, solutions to future water demand conflicts such as adjusting reservoir operations and building new reservoirs (Southam et al. 1999) may not. Protection of current levels of genetic diversity and the persistence of fragmented populations will require ensuring connectivity with other populations and by increasing the length or quality of available habitat (Hildebrand 2003; Jager 2006).

Acknowledgements The study was supported by Fisheries and Oceans Canada, Endangered Species Recovery Fund (WWF Canada and Environment Canada), Federal Interdepartmental Recovery Fund, Ontario Ministry of Natural Resources, and Industrial NSERC and Ontario Graduate scholarships awarded to S. Reid. Field sampling was assisted by P. Addison, J. Barnucz, A. Edwards, H. Gignac, N. Koutrilides and J. MacLeod. D. Gillette provided valuable assistance during laboratory data collection. Earlier versions of the manuscript were improved by comments provided by A. Dextrase and two anonymous reviewers.

Appendix

Appendix I Microsatellite allele size frequencies for Grand River black redhorse (*M. duquesnei*) populations

Locus	Allele size	CAL	CK	PAR	NR	GTO	GAL	CON	WM	All
RR55	108	0.14	0.23	0.15	0.13	0.07	0.11	0.17	0.14	0.14
	110	0.63	0.54	0.63	0.56	0.54	0.73	0.57	0.54	0.59
	112	0.01	0.01	0.00	0.00	0.01	0.00	0.00	0.02	0.01
	114	0.06	0.02	0.00	0.01	0.01	0.00	0.00	0.04	0.02
	138	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.00	0.00
	144	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	148	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00
	152	0.12	0.14	0.20	0.23	0.27	0.07	0.15	0.16	0.17
	154	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	164	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00
	166	0.00	0.00	0.00	0.02	0.01	0.00	0.07	0.02	0.01
	168	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	170	0.02	0.03	0.01	0.02	0.00	0.02	0.00	0.02	0.02
	172	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.01	0.00
	176	0.00	0.00	0.00	0.00	0.02	0.03	0.00	0.02	0.01
	178	0.00	0.00	0.00	0.00	0.01	0.02	0.00	0.00	0.00
	180	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
	182	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.00
	188	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.00
	190	0.01	0.00	0.01	0.01	0.04	0.00	0.00	0.01	0.01

Appendix I continued

Locus	Allele size	CAL	CK	PAR	NR	GTO	GAL	CON	WM	All
RR13	191	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	195	0.02	0.01	0.02	0.06	0.05	0.01	0.07	0.01	0.03
	201	0.68	0.61	0.64	0.69	0.52	0.57	0.48	0.56	0.60
	203	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	209	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	211	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00
	215	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	217	0.27	0.36	0.32	0.24	0.43	0.39	0.46	0.42	0.35
	219	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
	223	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	235	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	269	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
*dlu*4296	153	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00
	161	0.00	0.01	0.03	0.03	0.02	0.00	0.00	0.02	0.01
	165	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	169	0.02	0.03	0.02	0.00	0.04	0.03	0.02	0.07	0.03
	173	0.11	0.06	0.05	0.13	0.11	0.09	0.02	0.09	0.09
	177	0.12	0.16	0.14	0.13	0.13	0.13	0.15	0.16	0.14
	181	0.09	0.11	0.11	0.09	0.09	0.04	0.04	0.06	0.08
	185	0.11	0.11	0.12	0.09	0.15	0.19	0.15	0.12	0.13
	189	0.07	0.12	0.04	0.09	0.13	0.05	0.09	0.10	0.09
	193	0.05	0.07	0.11	0.11	0.06	0.04	0.09	0.09	0.08
	197	0.04	0.09	0.06	0.06	0.02	0.04	0.00	0.04	0.05
	201	0.06	0.02	0.03	0.11	0.05	0.06	0.02	0.02	0.05
	205	0.02	0.00	0.04	0.01	0.02	0.02	0.02	0.01	0.02
	209	0.01	0.01	0.01	0.02	0.04	0.01	0.02	0.01	0.02
	213	0.00	0.01	0.03	0.01	0.01	0.01	0.02	0.01	0.01
	217	0.02	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.01
	221	0.12	0.06	0.08	0.04	0.03	0.12	0.13	0.02	0.07
	225	0.08	0.04	0.04	0.03	0.05	0.02	0.02	0.06	0.05
	229	0.06	0.04	0.04	0.05	0.02	0.02	0.04	0.04	0.04
	233	0.02	0.02	0.03	0.01	0.01	0.05	0.11	0.04	0.03
	237	0.00	0.00	0.01	0.01	0.00	0.03	0.00	0.00	0.01
	241	0.01	0.01	0.02	0.00	0.00	0.02	0.04	0.01	0.01
	245	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
	249	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	253	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
*dlu*405	221	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.01	0.00
	225	0.00	0.01	0.04	0.02	0.02	0.02	0.00	0.02	0.02
	229	0.00	0.00	0.03	0.01	0.03	0.03	0.02	0.01	0.02
	233	0.00	0.05	0.09	0.01	0.02	0.10	0.02	0.07	0.05
	237	0.01	0.06	0.06	0.00	0.10	0.04	0.00	0.04	0.04
	241	0.01	0.02	0.04	0.00	0.04	0.04	0.00	0.07	0.03
	245	0.02	0.04	0.01	0.02	0.06	0.10	0.04	0.04	0.04
	249	0.02	0.01	0.00	0.00	0.02	0.00	0.00	0.01	0.01
	253	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	261	0.01	0.00	0.03	0.02	0.00	0.00	0.00	0.01	0.01
	265	0.00	0.00	0.01	0.02	0.01	0.00	0.00	0.00	0.01

Appendix I continued

Locus	Allele size	CAL	CK	PAR	NR	GTO	GAL	CON	WM	All
	269	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	273	0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	289	0.00	0.02	0.00	0.04	0.00	0.00	0.00	0.00	0.01
	293	0.03	0.01	0.03	0.03	0.01	0.00	0.02	0.01	0.02
	297	0.08	0.03	0.01	0.06	0.03	0.01	0.15	0.07	0.05
	301	0.08	0.06	0.04	0.07	0.06	0.06	0.04	0.06	0.06
	305	0.07	0.08	0.12	0.08	0.05	0.09	0.11	0.10	0.09
	309	0.12	0.10	0.06	0.11	0.08	0.04	0.09	0.09	0.09
	313	0.11	0.15	0.11	0.12	0.08	0.05	0.02	0.06	0.09
	317	0.15	0.16	0.11	0.19	0.12	0.12	0.20	0.12	0.14
	321	0.03	0.12	0.08	0.10	0.13	0.13	0.15	0.04	0.09
	325	0.15	0.02	0.06	0.06	0.04	0.04	0.11	0.11	0.07
	329	0.01	0.00	0.03	0.01	0.03	0.02	0.00	0.00	0.01
	333	0.08	0.01	0.01	0.05	0.03	0.06	0.02	0.05	0.04
	337	0.01	0.03	0.03	0.01	0.00	0.03	0.00	0.01	0.02
*dlu*45	273	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
	289	0.01	0.03	0.01	0.02	0.00	0.02	0.02	0.02	0.02
	293	0.00	0.00	0.01	0.01	0.01	0.03	0.00	0.01	0.01
	297	0.07	0.08	0.11	0.08	0.07	0.13	0.07	0.08	0.09
	301	0.03	0.08	0.05	0.07	0.07	0.08	0.11	0.05	0.07
	305	0.09	0.09	0.10	0.12	0.09	0.07	0.16	0.07	0.09
	309	0.46	0.36	0.41	0.28	0.48	0.34	0.27	0.40	0.38
	313	0.25	0.29	0.23	0.31	0.22	0.28	0.34	0.25	0.27
	317	0.08	0.04	0.06	0.06	0.04	0.03	0.02	0.10	0.06
	321	0.01	0.01	0.02	0.00	0.00	0.01	0.00	0.01	0.01
	325	0.00	0.02	0.00	0.01	0.00	0.00	0.00	0.00	0.00
	337	0.00	0.00	0.00	0.04	0.01	0.00	0.00	0.01	0.01
*Mohu-Lav*194	129	0.22	0.15	0.03	0.12	0.02	0.19	0.11	0.19	0.13
	133	0.29	0.39	0.45	0.39	0.62	0.15	0.24	0.34	0.37
	137	0.05	0.10	0.05	0.02	0.05	0.17	0.15	0.05	0.07
	141	0.02	0.04	0.03	0.06	0.01	0.04	0.04	0.03	0.03
	145	0.01	0.01	0.01	0.02	0.00	0.00	0.00	0.00	0.01
	149	0.00	0.01	0.02	0.04	0.01	0.01	0.04	0.01	0.02
	153	0.14	0.10	0.08	0.06	0.05	0.06	0.11	0.12	0.09
	157	0.04	0.02	0.08	0.05	0.01	0.04	0.02	0.04	0.04
	161	0.07	0.02	0.08	0.08	0.09	0.06	0.07	0.07	0.07
	165	0.13	0.11	0.14	0.15	0.12	0.23	0.22	0.13	0.15
	169	0.04	0.02	0.03	0.02	0.01	0.03	0.00	0.04	0.03
	173	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	181	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	229	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
*Mohu-Lav*296	116	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.00	0.00
	120	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.00	0.00
	124	0.00	0.01	0.20	0.10	0.00	0.03	0.00	0.06	0.05
	128	0.12	0.35	0.23	0.34	0.19	0.20	0.16	0.15	0.22
	132	0.00	0.00	0.00	0.02	0.00	0.00	0.02	0.00	0.00
	136	0.00	0.01	0.00	0.01	0.00	0.02	0.09	0.00	0.01
	140	0.01	0.02	0.00	0.01	0.00	0.00	0.02	0.00	0.01

Appendix I continued

Locus	Allele size	CAL	CK	PAR	NR	GTO	GAL	CON	WM	All
	144	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00
	148	0.07	0.00	0.00	0.02	0.00	0.09	0.05	0.13	0.05
	152	0.06	0.01	0.00	0.00	0.00	0.06	0.00	0.03	0.02
	156	0.00	0.00	0.02	0.01	0.01	0.00	0.05	0.00	0.01
	160	0.07	0.00	0.01	0.11	0.05	0.07	0.05	0.03	0.05
	164	0.07	0.16	0.04	0.01	0.17	0.07	0.00	0.10	0.08
	168	0.03	0.00	0.06	0.03	0.08	0.02	0.05	0.00	0.03
	172	0.10	0.10	0.12	0.09	0.07	0.05	0.09	0.08	0.09
	176	0.00	0.00	0.01	0.00	0.02	0.02	0.00	0.03	0.01
	180	0.12	0.14	0.01	0.05	0.00	0.08	0.09	0.17	0.08
	184	0.13	0.14	0.03	0.07	0.07	0.06	0.02	0.03	0.07
	188	0.05	0.02	0.07	0.04	0.06	0.19	0.09	0.15	0.08
	192	0.03	0.00	0.12	0.00	0.11	0.01	0.00	0.00	0.04
	196	0.02	0.00	0.05	0.03	0.07	0.00	0.05	0.00	0.03
	200	0.01	0.00	0.01	0.00	0.01	0.00	0.02	0.02	0.01
	204	0.00	0.00	0.01	0.02	0.01	0.00	0.00	0.00	0.01
	208	0.00	0.00	0.01	0.00	0.01	0.00	0.05	0.01	0.01
	212	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
	216	0.01	0.00	0.00	0.02	0.00	0.00	0.05	0.01	0.01
	220	0.01	0.01	0.02	0.00	0.04	0.00	0.00	0.01	0.01
	224	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	228	0.04	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01
	232	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	236	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00
	240	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	248	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
*Mohu-Lav*305	160	0.02	0.00	0.00	0.00	0.01	0.03	0.00	0.01	0.01
	164	0.11	0.13	0.06	0.09	0.09	0.06	0.17	0.15	0.10
	168	0.11	0.08	0.07	0.15	0.07	0.06	0.02	0.03	0.08
	172	0.61	0.72	0.74	0.66	0.70	0.56	0.70	0.64	0.66
	176	0.01	0.00	0.02	0.02	0.00	0.05	0.04	0.02	0.02
	180	0.03	0.03	0.00	0.01	0.00	0.05	0.02	0.03	0.02
	184	0.02	0.01	0.00	0.00	0.01	0.03	0.00	0.02	0.01
	188	0.03	0.00	0.00	0.00	0.00	0.02	0.00	0.03	0.01
	192	0.05	0.01	0.11	0.08	0.11	0.12	0.04	0.08	0.08
	196	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
	200	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	204	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
	236	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00

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